

RE 497/597 A: Real Estate Data Modeling

Instructor: Shiqi Ding (shiqidi@uw.edu)

Classroom: Gould Hall 007F

Class Days and Time: Tuesdays and Thursdays, 12:30 - 2:20 PM

Office Hours: Tuesdays: 2:20 - 3:20 PM, and Thursdays: 2:20 – 3:00 PM (see Canvas for more details)

Course Description: This course will cover spatial analysis of real estate. The course will teach students methods in quantitative analysis of spatial data and how to collect, process, analyze, manage, and model a variety of data sets to make real estate decisions. Suitability and exploratory data analyses will be covered.

Course Objectives: There are three main objectives in this course: 1) examine real estate problems from a quantitative perspective; 2) describe and analyze data and variable relationships using a variety of spatial techniques; and 3) interpret and communicate the results to various audiences.

Textbooks and Readings: There are no required textbooks for the course. All the materials for the course are available online and can be accessed via Canvas.

Project-Based Course: This is a project-based course where students will conduct a suitability analysis. Esri applications and tools such as ArcGIS Pro and Community Analyst will be used to develop the suitability model. The purpose of the project is to learn how to use data for decision-making of real property acquisition by identifying the best locations for real estate assets. Students will form teams of three or four (to be decided) to work on the project and assignments. The project process involves gathering data, learning about different data sets and data types, processing and preparing data for analyses, understanding the data with basic descriptive statistics and spatial analysis, standardizing and transforming data, and weighting and combining the data to build a suitability model.

Course Structure:

By 12 PM Before Class	Complete module quiz
12:30 PM – 12:45 PM	Module and Project Q & A session (attendance highly encouraged)
12:45 PM – 1:30 PM	Discuss and demo group assignment (attendance required for assignment grade)
1:30 PM – 2:20 PM	In-class group assignment (attendance required for assignment grade)
After Class	Complete all readings, web courses, and PowerPoints for next module

Course Requirements: The grade for the course is based on quizzes, group assignments, and a final presentation.

- 1. Quizzes:** There is a short 15-question quiz for every module that covers the required readings, web courses, and PowerPoints. The quiz must be taken on Canvas, and you will be given 30 minutes to complete the quiz. The quiz should be taken after you have completed reviewing the resources for each module. The quiz for each module is available as soon as the prior class ends and up to 30 minutes before the module will be discussed in class. The quiz must be taken in one sitting, and you must complete the quiz on your own. You may not share the quiz questions and answers. There are no make-up quizzes allowed since you may take the quiz at any time between modules and only the highest 15 quiz scores will be counted toward your grade.
- 2. Group Assignments:** Students will work on group assignments in class. Each assignment builds on a component of the project. The instructions for the group assignments will start promptly at 1:00 PM. If you arrive after 1 PM or miss class, or leave before your group finishes the assignment, you will receive zero points for the assignment. Students will be able to complete several assignments during class, but

the assignments aren't due until a later date to provide more time out of class to complete the assignment if needed. There are no make-up assignments allowed since only the highest 15 scores will be counted toward your grade.

- 3. Final Poster and Presentation:** Plan to prepare a poster and a 10-minute talk about your suitability model at the end of the course (7 minutes for presentation and 3 minutes for Q&A). Both undergraduate and graduate students are responsible for preparing the poster. Graduate students will be responsible for presenting the poster. **Important:** Your poster should be in PDF format, 24" x 36", using the UW research poster templates (<https://www.washington.edu/brand/templates/presentations/>). **Q&A responsibility:** Each group is required to ask at least one question throughout all presentations (to any group they choose, but not to every group). All students are responsible for answering questions from other groups. The poster and talk should include: an overview of your project and project criteria, describe the data and variables used in your model and explain why, discuss the transformation of your data and any weights applied, and discuss the different sensitivity tests performed. Also, identify the top 10 sites selected and present one in detail. The grading criteria for the presentation and project are outlined in the rubric. Any student that is not present for the presentation day will automatically have 10 points deducted from their final grade. Each group should submit the poster to Canvas as a PDF file by **5:00 PM on Tuesday, December 10th**.

Grading Rubric for Suitability Model Final Project and Presentation:

Criteria	Exemplary (A)	Acceptable (B)	Fair (C)	Unacceptable
Poster Content and Presentation of the Model	9 or 10 Points The model is demonstrated so the audience can easily understand its purpose and use and includes all the project components	8 Points The model is demonstrated so the audience can understand its purpose and use but more information is needed to understand the full scope	7 Points The model is demonstrated but there are some components regarding the purpose and use that are unclear or missing	6 or Less Points The presentation of the model lacks clarity, and its purpose and use are unclear, or components are missing
Poster Design	5 Points The overall design and layout of the poster is appealing, the maps and data are easy to read and understand with a color/pattern scheme and labels that are appropriate	4 Points Effort is seen in the overall design and layout of the poster, but some improvement is needed to make the overall design more appealing	3 Points The overall design and layout of the poster needs improvement to make the maps and data easier to read and changes to labels and color/pattern scheme are needed	2 or Less Points The overall design and layout are poor and there is no discernible design theme
Organization and Writing	4 Points The model is well organized with no grammar or spelling errors	3 Points The model is organized but there are a few writing errors	2 Points Little discernible organization and several writing errors	1 or Less Points No organization and significant writing errors
Peer-Questions	3 Points Asked at least one insightful question that demonstrated a good understanding of the project or invited a discussion.	2 Points Asked one relevant question that demonstrated a good understanding of the project.	1 Points Asked one question that is unclear or irrelevant.	0 Points Asked no questions.

Time	3 Points Presentation is given in 7 minutes	2 Points Presentation is over/under by 2 minutes	1 Points Presentation is over/under by 3 minutes	0 Points Presentation is over/under by more than 5 minutes
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Course Evaluation:

	Number of Points	Total Number	Total Points
Quizzes	15	15	225 points
Group Assignments	10	15	150 points
Poster and Lightning Talk	25	1	25 points
		Total Points:	400 points

Late Policy:

- 1. Quizzes and Final Presentation:** No late submissions will be accepted.
- 2. Group Assignments:** Each assignment will list its due date. Each student will have a **total of five “free” late days** (a late day is 24 hours of lateness). There are no partial days, so assignments are either on time, 1 day late, 2 days late, etc. Because of this generous late policy, students will not be granted extensions for assignments unless they have some highly extenuating circumstances. Once a student has used up all of their late days, each successive late day will result in a loss of 1 point. All assignments must be submitted **by 5:00 PM of the last day of class (Thursday, December 5th)**, whether or not a student has free late days left.

Gradebook: The gradebook will be updated weekly in Canvas. Please refer to the gradebook frequently to verify the number of points you have accumulated and your current standing in the course. Please notify me immediately if you believe an error has been made.

Contingency Plan: If there is an unexpected disruption during the quarter that results in a university mandate of canceled classes and/or a truncated quarter, your grade will be calculated based on the total points you have earned out of the total points possible up to the date of closure. Therefore, it is important to stay on track for all coursework and not depend on future assignments to improve your grade. This policy will only go into effect if there is a mandated university-wide cancellation of the remainder of the quarter.

Undergraduate Grading Scale:

A	4.0	94% and above	376+ points
	3.9	93%	372 – 375 points
A-	3.8	92%	368 – 371 points
	3.7	91%	364 – 367 points
	3.6	90%	360 – 363 points
	3.5	89%	356 – 359 points
B+	3.4	88%	352 – 355 points
	3.3	87%	348 – 351 points
	3.2	86%	344 – 347 points
B	3.1	85%	340 – 343 points
	3.0	84%	336 – 339 points
	2.9	83%	332 – 335 points
B-	2.8	82%	328 – 331 points
	2.7	81%	324 – 327 points

	2.6	80%	320 – 323 points
	2.5	79%	316 – 319 points
C+	2.4	78%	312 – 315 points
	2.3	77%	308 – 311 points
	2.2	76%	304 – 307 points
C	2.1	75%	300 – 303 points
	2.0	74%	296 – 299 points
	1.9	73%	292 – 295 points
C-	1.8	72%	288 – 291 points
	1.7	71%	284 – 287 points
	1.6	70%	280 – 283 points
	1.5	69%	276 – 279 points
D+	1.4	68%	272 – 275 points
	1.3	67%	268 – 271 points
	1.2	66%	264 – 267 points
D	1.1	65%	260 – 263 points
	1.0	64%	256 – 259 points
	0.9	63%	252 – 255 points
D-	0.8	62%	248 – 251 points
	0.7	61%	244 – 247 points
E	0.0	60% and below	243 points and below

Graduate Grading Scale:

A	4.0	94% and above	376+ points
	3.9	93%	372 – 375 points
A-	3.8	92%	368 – 371 points
	3.7	91%	364 – 367 points
	3.6	90%	360 – 363 points
	3.5	89%	356 – 359 points
B+	3.4	88%	352 – 355 points
	3.3	87%	348 – 351 points
	3.2	86%	344 – 347 points
	3.1	85%	340 – 343 points
B	3.0	84%	336 – 339 points
	2.9	83%	332 – 335 points
B-	2.8	82%	328 – 331 points
	2.7	81%	324 – 327 points
	2.6	80%	320 – 323 points
	2.5	79%	316 – 319 points
C+	2.4	78%	312 – 315 points
	2.3	77%	308 – 311 points
	2.2	76%	304 – 307 points
	2.1	75%	300 – 303 points
C	2.0	74%	296 – 299 points
	1.9	73%	292 – 295 points
C-	1.8	72%	288 – 291 points
	1.7	71%	284 – 287 points
E	1.6	70% and below	283 points and below

Class Attendance Policy: Attending class in person is mandatory and is strongly encouraged. You will only benefit from the discussions and work on the assignments if you are present for the in-person sessions.

Classroom Etiquette:

1. Be punctual for all classes.
2. Be respectful of the beliefs of others.
3. Turn off cell phones and other electronic devices that may be disruptive to class.
4. If you need to leave the room, be as quiet as possible when entering and exiting the class while in session.
5. The use of any recording device without permission is prohibited.

Meeting Protocol for the Virtual Classroom (if necessary): The class will be taught in person at Gould 007F. This section is only applicable if the course moves online during the quarter. Zoom is integrated into Canvas and you should be able to connect to each class via Canvas. Please test all technology before class, including your video, audio, and Wi-Fi connection. Make sure your camera, mic, and speakers are all working properly. Choose an appropriate space for joining the virtual classroom (a quiet place where you can focus and will not be disturbed). If there is a technical issue with Zoom, please watch the Canvas announcements as I will post instructions on how we will handle the technical issue.

During class, please follow these guidelines:

- Muting and unmuting your mic: Your microphone is set on mute when you enter the virtual classroom. Please unmute yourself to speak. Mute your mic when you are not speaking to eliminate ambient feedback and other unwanted background noises.
- Start Video: Your video is off by default at the start of the class. Please start your video when class starts and keep the camera on during the entire class, especially when you are speaking. For class engagement, it is important that we can see each other. If you are interrupted during class or need to take a quick break to use the restroom, please turn off your video but restart it as soon as you return.
- Chat: We will not be using this function in this course since it can be distracting. If you have questions or comments, please unmute your mic or raise your hand.
- Raise Hand: During class, you can use the Raise Hand button to speak.
- Share Screen: When presenting using a PowerPoint or another application, share the application only rather than your desktop to protect your privacy.

Academic Integrity: Academic integrity is an important value of this course. Students engaging in academic misconduct, particularly plagiarism, will be referred to the Dean's office for further investigation. VeriCite is used in Canvas for all written assignments and plagiarism will not be tolerated in this course. If you use more than 5 words in a row from another document without quotation marks, even if you include the source in parentheses, this is considered plagiarism. Exact language must be in quotes and there are very few situations where quotes are acceptable. I must report plagiarism and it will result in an investigation that can lead to disciplinary sanctions. Please read the [UW Student Conduct Code](#) with special attention to the [Academic Misconduct section](#).

Disability Accommodations: The Runstad Department of Real Estate is committed to ensuring learning opportunities for students with disabilities. If you would like to request academic accommodations due to a disability, please contact the [Office of Disability Resources for Students](#). If you have a letter from the Office of Disability Resources for Students indicating you have a disability that requires academic accommodations, please present it to me as soon as possible so we can discuss specific accommodations for this course.

Religious Accommodations Policy: Washington state law requires that UW develop a policy for accommodation of student absences or significant hardship due to reasons of faith or conscience, or for

organized religious activities. The UW's policy, including more information about how to request an accommodation, is available at [Religious Accommodations Policy](#). Accommodations must be requested within the first two weeks of this course using the [Religious Accommodations Request form](#).

Student Athletes: Student athletes who have conflicts with attending any lecture or submitting any assignment should submit a list of such conflicts along with a letter from the athletics department by Tuesday, April 2, 2024. I will reasonably accommodate any student athlete who follows this guideline.

Safety: Students are advised to refer to UW policies and procedures to ensure their safety and security on campus. For more information, go to: <http://www.washington.edu/safecampus/>. To report threats, seek advice, or get counseling, dial 206-685-SAFE (7233).

ArcGIS Online: Each student will be given access to ArcGIS Online. ArcGIS Online can only be used for academic purposes related to your course work. For all Esri products, you will sign in **through the website** and use the **ENTERPRISE LOGIN**. The ArcGIS organization's URL is [uwsupplemental.maps.arcgis.com](https://www.esri.com/arcgis/enterprise). You will then continue to sign in using your UW NetID and UW Password. At the end of the quarter, your ArcGIS Online account will be deleted as well as any materials that you have stored in your online account. If you wish to have your account active until you graduate, you must notify me in writing before the end of the quarter. If you request extended access, your account and all related materials will be deleted when you graduate. You can move content in your account to other accounts using the ArcGIS Online Assistant: <https://ago-assistant.esri.com/>. At the end of the quarter please delete all your groups and content.

ArcGIS Pro: Each student will be given access to the ArcGIS Pro software. ArcGIS Pro can only be used for academic purposes related to your course work. **Important:** Using some geoprocessing tools in ArcGIS Pro and Esri Community Analyst may incur credit usage against your AGOL account.

Suzzallo Library GIS Lab: The Suzzallo Library has Geospatial Data Resources and a GIS lab. The lab also has staff that can provide assistance with basic GIS tasks. Visit the following webpage for the hours that the GIS assistant is available: <https://guides.lib.uw.edu/research/gis/lab>

Esri Resources:

ArcGIS Online and ArcGIS Pro

ArcGIS Online Sign In (<https://www.arcgis.com/home/signin.html>)

ArcGIS Online Help (<https://doc.arcgis.com/en/arcgis-online/>)

ArcGIS Online Credits (<http://www.esri.com/software/arcgis/arcgisonline/credits>)

ArcGIS Pro Help (<https://pro.arcgis.com/en/pro-app/help/main/welcome-to-the-arcgis-pro-app-help.htm>)

Learning Resources

Esri Training (<https://www.esri.com/training/>)

Applied Analysis (<http://desktop.arcgis.com/en/analytics/casestudies/>)

GeoNet The ESRI Community (<https://community.esri.com/>)

Learn ArcGIS (<https://learn.arcgis.com/en/>)

ArcGIS for Developers (<https://developers.arcgis.com/>)

Esri User Conference (<https://www.esri.com/en-us/about/events/uc/overview>)

Class and Due Dates	Topics
September 26, 2024 Introduction:	Welcome and Syllabus Overview Esri Overview and ArcGIS Pro/ArcGIS Online Account Setup
October 1, 2024 Quiz 1: Module 1: Assignment:	Due by 12:00 PM Introduction to Suitability Modeling Group Assignment 1 due by October 2, 2024, at 12:00 PM
October 3, 2024 Quiz 2: Module 2: Assignment:	Due by 12:00 PM Getting Started with ArcGIS Pro Group Assignment 2 due by October 7, 2024, at 12:00 PM
October 8, 2024 Quiz 3: Module 3: Assignment:	Due by 12:00 PM Intro to Spatial Data, Geodatabases, and Data Management Group Assignment 3 due by October 9, 2024, at 12:00 PM
October 10, 2024 Quiz 4: Module 4: Assignment:	Due by 12:00 PM Datasets and Data Sources Group Assignment 4 due by October 14, 2024, at 12:00 PM
October 15, 2024 Quiz 5: Module 5: Assignment:	Due by 12:00 PM Geoprocessing and Editing Group Assignment 5 due by October 16, 2024, at 12:00 PM
October 17, 2024 Quiz 6: Module 6: Assignment:	Due by 12:00 PM Coordinate Systems and Projections Group Assignment 6 due by October 21, 2024, at 12:00 PM
October 22, 2024 Quiz 7: Module 7: Assignment:	Due by 12:00 PM Geocoding and Creating XY Event Layers Group Assignment 7 due by October 23, 2024, at 12:00 PM
October 24, 2024 Quiz 8: Module 8: Assignment:	Due by 12:00 PM Spatial and Table Joins Group Assignment 8 due by October 28, 2024, at 12:00 PM
October 29, 2024 Quiz 9: Module 9: Assignment:	Due by 12:00 PM Descriptive Statistics Group Assignment 9 due by October 30, 2024, at 12:00 PM
October 31, 2024 Quiz 10: Module 10: Assignment:	Due by 12:00 PM Correlation Group Assignment 10 due by November 4, 2024, at 12:00 PM
November 5, 2024 Quiz 11: Module 11: Assignment:	Due by 12:00 PM Basics of Cartography Group Assignment 11 due by November 6, 2024, at 12:00 PM

November 7, 2024	No class. The instructor is attending a conference.
November 12, 2024 Quiz 12: Module 12: Assignment:	Due by 12:00 PM Analyzing Spatial Patterns Group Assignment 12 due by November 13, 2024, at 12:00 PM
November 14, 2024 Quiz 13: Module 13: Assignment:	Due by 12:00 PM Identifying Clusters Group Assignment 13 due by November 18, 2024, at 12:00 PM
November 19, 2024 Quiz 14: Module 14: Assignment:	Due by 12:00 PM Standardizing and Transforming Data Group Assignment 14 due by November 20, 2024, at 12:00 PM
November 21, 2024 Quiz 15: Module 15: Assignment:	Due by 12:00 PM Weight Data and Create Index Group Assignment 15 due by November 25, 2024, at 12:00 PM
November 26, 2024 Quiz 16: Module 16: Assignment:	Due by 12:00 PM Performing Sensitivity Analysis Group Assignment 16 due by November 27, 2024, at 12:00 PM
November 28, 2024	Thanksgiving Break
December 3, 2024 Quiz 17: Module 17: Assignment:	Due by 12:00 PM Overlays and Locating Top Sites Group Assignment 17 due by December 4, 2024, at 12:00 PM
December 5, 2024	Final Posters Presentation PDF due by December 10, 2024, at 5:00 PM